Warmup · Two Questions

Start here. Both questions review skills from Sections 3.6 and 3.7; inverse trigonometric derivatives come later in today's lesson.

Warmup 1 · Recent review

No Calculator

$x^2 + xy + y^2 = 7$ passes through $(1, 2)$. What is the slope of the tangent line at that point?

A. $-\frac{5}{4}$

B. $-\frac{4}{5}$

C. $-\frac{1}{5}$

D. $\frac{4}{5}$

E. $\frac{5}{4}$

Warmup 2 · Spiral review

Calculator Permitted

If $h(x) = \sin(2x^2 - x)$, what is $h'(0.7)$ to the nearest thousandth? Angles are in radians.

A. 0.495

B. 1.350

C. 1.730

D. 1.800

E. 2.497

Name _____ Date _____

Use the workspace beneath each question to justify your choice, not just circle a letter.

Learning Focus & Vocabulary

Learning focus

By the end of this lesson, I can …

- Differentiate inverse trigonometric functions using the correct rule and the Chain Rule.
- Explain why the inverse sine derivative follows from implicit differentiation.
- Find the derivative of an inverse function at a point without solving for its inverse.
- Recognize when a derivative is undefined or a rule requires a domain restriction.

Vocabulary to highlight

inverse function
Undo a one-to-one function; inputs and outputs switch.

inverse trig
An angle returned from a trig ratio, with restricted range.

principal range
Chosen output interval making the inverse single-valued.

arcsin / arctan
Notation for inverse sine and inverse tangent.

reciprocal slopes
Derivatives of inverses at corresponding points.

implicit differentiation
Differentiate a relationship without isolating a variable.

Why the new rules make sense

Inverse trig functions answer a question such as “Which angle has sine equal to this input?” The input is a ratio; the output is an angle in radians.

For $y = \arcsin x$, rewrite as $\sin y = x$. Because y depends on x, differentiating the left side requires the Chain Rule:

$$\cos y \, \frac{dy}{dx} = 1$$

On the principal range of arcsine, $\cos y = \sqrt{1 - \sin^2 y} = \sqrt{1 - x^2}$. Therefore the inverse-sine rule follows.

Stop & reflect: Why is $\cos y$ positive in that last step? What range of outputs does $\arcsin x$ use?

Rules & Inverse Slopes

Six inverse-trig derivative rules

In each rule, $u = u(x)$. The last factor comes from the Chain Rule. Square-root expressions require real values and nonzero denominators.

Inverse sine

$$\frac{d}{dx}\arcsin u = \frac{u\prime}{\sqrt{1-u^2}}$$

Inverse cosine

$$\frac{d}{dx}\arccos u = -\frac{u\prime}{\sqrt{1-u^2}}$$

Inverse tangent

$$\frac{d}{dx}\arctan u = \frac{u\prime}{1+u^2}$$

Inverse cotangent

$$\frac{d}{dx}\operatorname{arccot} u = -\frac{u\prime}{1+u^2}$$

Inverse secant

$$\frac{d}{dx}\operatorname{arcsec} u = \frac{u\prime}{|u|\sqrt{u^2-1}}$$

Inverse cosecant

$$\frac{d}{dx}\operatorname{arccsc} u = -\frac{u\prime}{|u|\sqrt{u^2-1}}$$

Notation check: $\sin^{-1} x = \arcsin x$ denotes an inverse trig function, **not** $\frac{1}{\sin x}$. Likewise, $\arccos$ and arcsec are different functions.

Derivative of an inverse at a point

Swap coordinates; take the reciprocal

If $f(a) = b$ and $f\prime(a) \neq 0$, the inverse passes through (b, a) and

$$(f^{-1})\prime(b) = \frac{1}{f\prime(a)}$$

Procedure: solve $f(a) = b$ for the original input a, evaluate $f\prime(a)$, then take its reciprocal. **Do not** evaluate $f\prime(b)$ unless $a = b$.

Examples 1–2 · Inverse Sine & Cosine

These are the six examples from the supplied Section 3.8 notes. After each example, complete a closely related **Your Try** independently.

Example 1 · Source — **Source example**

Suppose $y = \arcsin x$. Find $\frac{dy}{dx}$ using **implicit differentiation**.

Your Try 1 — **Your Try**

Let $y = \arccos x$. Start with $\cos y = x$ and derive $\frac{dy}{dx}$.

Example 2 · Source — **Source example**

Find the derivative of $f(t) = \arcsin(t^2)$. Identify $u(t)$ and $u\prime(t)$.

Your Try 2 — **Your Try**

Find $\frac{dy}{dt}$ for $y = \arccos(3t)$. State where your derivative is defined.

Examples 3–4 · Chain Rule & Inverses

Example 3 · Source — **Source example**

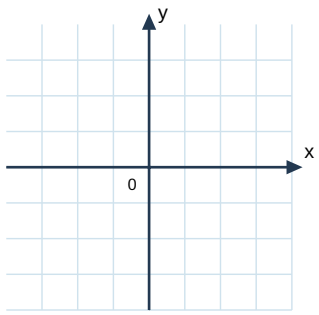
Find the derivative of $y = \arctan(\sqrt{x-1})$.

Your Try 3 — **Your Try**

Differentiate $y = \arctan(\sqrt{x+2})$.

Example 4 · Graphing a function and its inverse

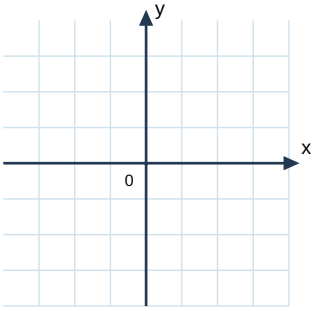
For $y = 4x + 1$: (a) find its slope; (b) graph the inverse; (c) find the inverse slope; (d) map the point $(2, 9)$ to the inverse.



Graph both lines and the line of reflection $y = x$.

Your Try 4

Repeat (a)–(d) for $y = 3x - 2$, using the original point $(2, 4)$.



Examples 5–6 · Inverse Derivatives

For each inverse-at-a-point problem, show the matching point on f, evaluate the original derivative there, and take the reciprocal.

Example 5 · Source — **Source example**

Let $f(x) = x^5 + 2x - 1$. Verify that $(0, -1)$ lies on the graph. Find $(f^{-1})\prime(-1)$.

Your Try 5 — **Your Try**

Let $f(x) = x^3 + 4x + 2$. Verify $(0, 2)$ and find $(f^{-1})\prime(2)$.

Example 6 · Source — **Source example**

Let $f(x) = x^3 + 2x - 1$. Find $(f^{-1})\prime(2)$ without finding an inverse formula.

Your Try 6 — **Your Try**

Let $f(x) = x^3 + 3x - 1$. Find $(f^{-1})\prime(3)$.

Practice · Exercises 1–3

Independent practice · Show your reasoning and simplify when appropriate. These are new problems, not repeats of the source examples.

1. Differentiate $y = \arcsin(3x - 1)$.

Use the inverse-sine rule and the Chain Rule.

2. Differentiate $h(x) = x \arctan x$.

Two distinct differentiation rules are needed.

3. Find $\frac{dy}{dx}$ if $y = \arccos(x^2)$.

Your answer can use a square root.

Practice · Exercises 4–6

Independent practice · Show your reasoning and simplify when appropriate. These are new problems, not repeats of the source examples.

4. Find $\frac{dy}{dx}$ if $y = \operatorname{arcsec}(2x)$.

Pay attention to absolute value and the domain.

5. Differentiate $g(x) = \frac{\arctan x}{x}$, for $x \neq 0$.

Use the quotient rule with an inverse-trig function.

6. Let $f(x) = x^3 + 4x - 2$. Find $(f^{-1})'(3)$.

Solve for the original input first; do not invert the cubic algebraically.